

TECHNICAL DOCUMENTATION

# EngineerTutorial

Complete System Design & Architecture Reference

Next.js 14 App Router

MongoDB Atlas

NextAuth.js

Razorpay

Judge0 API

Tailwind CSS

AI Chatbot

Block CMS

**Stack:** Next.js 14 App Router · MongoDB Atlas · NextAuth.js · Razorpay · Monaco Editor · Judge0 · Tailwind CSS  
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## 1. Project Overview

### 1.1 What We Are Building

A production-grade CS learning platform similar to **GeeksforGeeks** and **HelloInterview**, focused on:

- System Design, Apache Kafka, LLD, HLD, AWS, GenAI, DSA
- Built-in Monaco code editor with Judge0 execution
- AI chatbot sidebar (proxied to a separate AI service)
- Free + Pro subscription model (Razorpay)
- Fully CMS-driven admin panel (no code deploy to change content)

### 1.2 Key Differentiators

| Differentiator             | Description                                                                                       |
|----------------------------|---------------------------------------------------------------------------------------------------|
| Block-based CMS            | All content is <code>blocks[]</code> JSON — admin edits without touching code                     |
| Server-side content gating | API slices <code>blocks[]</code> for free users — security is NOT on the client                   |
| Context-aware AI           | AI chatbot auto-injects the current topic into its system prompt                                  |
| DB-driven navigation       | Header, footer, announcement bar all in MongoDB — live changes, zero deploy                       |
| Dual image upload          | Admin can upload to Cloudinary (CDN URL) OR static <code>/public</code> folder (self-hosted path) |
| All blogs free             | <code>isPremium</code> flag exists ONLY on Topics — every blog is publicly readable               |

### 1.3 User Roles & Access Matrix

| Role                      | Free Topics | Premium Topics            | Blogs  | Dashboard | Admin Panel             |
|---------------------------|-------------|---------------------------|--------|-----------|-------------------------|
| <code>guest</code>        | ✓ Full      | 🔒 Preview only            | ✓ Full | ✗         | ✗                       |
| <code>free_user</code>    | ✓ Full      | 🔒 Preview + upgrade modal | ✓ Full | ✓         | ✗                       |
| <code>premium_user</code> | ✓ Full      | ✓ Full                    | ✓ Full | ✓         | ✗                       |
| <code>author</code>       | ✓ Full      | ✓ Full                    | ✓ Full | ✓         | ✓ Content only          |
| <code>admin</code>        | ✓ Full      | ✓ Full                    | ✓ Full | ✓         | ✓ Full                  |
| <code>super_admin</code>  | ✓ Full      | ✓ Full                    | ✓ Full | ✓         | ✓ Full + promote others |

## 2. Functional Requirements

### 2.1 User Management

#### 2.1.1 Authentication

| ID     | Requirement                                                    | Priority |
|--------|----------------------------------------------------------------|----------|
| FR-1.1 | Register with email + password (bcrypt hashed)                 | P0       |
| FR-1.2 | Login with Google OAuth (NextAuth GoogleProvider)              | P0       |
| FR-1.3 | Login with GitHub OAuth (NextAuth GithubProvider)              | P0       |
| FR-1.4 | JWT-based session — role + subscriptionStatus in token payload | P0       |
| FR-1.5 | Email verification required after email/password registration  | P1       |
| FR-1.6 | Password reset via OTP email link                              | P1       |
| FR-1.7 | "Remember me" — 30-day token expiry vs default 24h             | P2       |
| FR-1.8 | Account deletion — full data wipe (User, Progress, Bookmarks)  | P2       |

#### 2.1.2 User Profile

| ID     | Requirement                                                                         |
|--------|-------------------------------------------------------------------------------------|
| FR-2.1 | Display name, bio, avatar image URL                                                 |
| FR-2.2 | Link GitHub and LinkedIn profile URLs                                               |
| FR-2.3 | Preferred theme (light / dark / system) — persisted in DB and synced across devices |
| FR-2.4 | Toggle email notification preferences per type                                      |
| FR-2.5 | Subscription status badge shown on profile                                          |

### 2.1.3 Progress Tracking

| ID     | Requirement                                                       |
|--------|-------------------------------------------------------------------|
| FR-3.1 | "Mark as Complete" button on every topic                          |
| FR-3.2 | Per-course completion percentage (0–100%)                         |
| FR-3.3 | Course sidebar shows green checkmark on completed topics          |
| FR-3.4 | "Resume Learning" button — links to last visited topic per course |
| FR-3.5 | User dashboard shows all in-progress courses with progress bars   |
| FR-3.6 | Daily login streak counter (consecutive active days)              |

### 2.1.4 Bookmarks

| ID     | Requirement                                                      |
|--------|------------------------------------------------------------------|
| FR-4.1 | Bookmark any topic or blog post with one click                   |
| FR-4.2 | Bookmarks tab in dashboard lists all saved items (topic or blog) |
| FR-4.3 | Remove bookmarks                                                 |

## 2.2 Content Management

### 2.2.1 Content Hierarchy

```

Platform
├─ Course      (e.g., "System Design")
│  └─ Section  (e.g., "Fundamentals")
│     └─ Topic  (e.g., "What is Load Balancing?") ← can be isPremium
│     └─ Topic  (e.g., "CAP Theorem")
│  └─ Section  (e.g., "Advanced Topics")
│     └─ Topic  ...
└─ Blog        (e.g., "Why Netflix Never Crashes") ← ALWAYS FREE
  
```

### 2.2.2 Block-Based Content Format

Every Topic and Blog stores content as an **ordered array of blocks**. Each block: `{ type: string, content: any }`.

Block Type

content

Rendered As

heading

{ text }

— main section title

smallHeading

{ text }

— subsection

mediumHeading

{ text }

— sub-subsection

text

{ text }

— supports **bold**, *italic*, ``[[label\``

list

{ items: string[] }

bullet list

orderedList

{ items: string[] }

numbered list

code

{ language, code, filename? }

Syntax-highlighted code block + Copy + Run button

table

{ headers: string[], rows: string[] [] }

image

{ src, alt, caption? }

with lazy loading

callout

`{ type: 'info'\`

'warning'\`

'tip',}`

video

`{ url, type: 'youtube'\`

'hosted' }`

Embed player

quiz

{ question, options: string[], answer: number }

MCQ with reveal

badge

{ items: {label, color}[] }

Tag chips row

line

{ }

divider

gap

`{ size: 'sm'\`

'md'\`

'lg'

Block Type

content

Shape

Rendered As

2.2.3 Topic Requirements

| ID     | Requirement                                                                                                                   |
|--------|-------------------------------------------------------------------------------------------------------------------------------|
| FR-5.1 | Topics have: slug, title, heading, blocks[], isPremium, previewBlockCount[], status                                           |
| FR-5.2 | <code>isPremium: true</code> enables content gating. Default: false                                                           |
| FR-5.3 | <code>previewBlockCount</code> - number of blocks shown to free users. Default: 3                                             |
| FR-5.4 | Status: <code>draft</code> → <code>published</code> → <code>archived</code> . Only <code>published</code> served to end users |
| FR-5.5 | Topics have full SEO fields: metaTitle, metaDescription, ogImage, canonicalUrl                                                |
| FR-5.6 | "Scheduled publish" - admin sets future datetime; cron or revalidation                                                        |

2.2.4 Blog Requirements

| ID     | Requirement                                                                              |
|--------|------------------------------------------------------------------------------------------|
| FR-6.1 | Blogs use identical block format to Topics                                               |
| FR-6.2 | <b>NO</b> <code>isPremium</code> field on Blogs - all blogs are always publicly readable |
| FR-6.3 | Blogs have: slug, title, subtitle, coverImage, tags[], author, publishedAt               |
| FR-6.4 | Related articles suggested at bottom (matched by tags)                                   |
| FR-6.5 | Blog listing shows cover image, tags, estimated read time, rating                        |

2.2.5 Code Block Features

| ID     | Requirement                                                                       |
|--------|-----------------------------------------------------------------------------------|
| FR-7.1 | Syntax highlighting on all code blocks (read-only in content view)                |
| FR-7.2 | "Copy Code" button on every code block                                            |
| FR-7.3 | "Open in Playground" button - loads code + language into <code>/playground</code> |
| FR-7.4 | "Ask AI about this code" button - injects code into AI chat window                |
| FR-7.5 | Multiple language tabs per code block (e.g., Python / Java / Go)                  |

Block Type

content  
Shape

Rendered As

## 2.3 Content Gating

| Scenario                  | API Returns                                                            | Client       |
|---------------------------|------------------------------------------------------------------------|--------------|
| Guest → free topic        | All blocks                                                             | Full         |
| Guest → premium topic     | <code>blocks[0..previewBlockCount]</code> + <code>isGated: true</code> | Preview      |
| Free user → premium topic | <code>blocks[0..previewBlockCount]</code> + <code>isGated: true</code> | Preview mode |
| Pro user → any topic      | All blocks + <code>isGated: false</code>                               | Full         |
| Any user → any blog       | All blocks                                                             | Full         |

**Security Note:** The API enforces gating. The client blur overlay is UX-only overlay via DevTools still only has the preview blocks – there are no additional

## 2.4 Code Editor & Execution

| ID      | Requirement                                                                   |
|---------|-------------------------------------------------------------------------------|
| FR-8.1  | Monaco Editor ( <code>@monaco-editor/react</code> ) – VS Code engine          |
| FR-8.2  | 8 languages: JavaScript, Python, Java, C++, Go, SQL, Rust, Bash               |
| FR-8.3  | Theme auto-syncs with site light/dark toggle                                  |
| FR-8.4  | Font size +/- controls                                                        |
| FR-8.5  | "Reset to starter code" restores original block content                       |
| FR-8.6  | Code execution via Judge0 API (sandboxed – never on own servers)              |
| FR-8.7  | Execution: 5s timeout, 256MB memory limit                                     |
| FR-8.8  | Output panel: stdout, stderr, execution time (ms), memory (KB), exit code     |
| FR-8.9  | stdin input field for programs requiring user input                           |
| FR-8.10 | Rate limiting: 10 runs/minute per IP (Node.js in-memory Map – reset on login) |
| FR-8.11 | Standalone <code>/playground</code> page with full-screen layout              |
| FR-8.12 | <code>?lang=python</code> query param pre-selects language in playground      |

Block Type

content

Shape

Rendered As

## 2.5 AI Chatbot

| ID      | Requirement                                                                    |
|---------|--------------------------------------------------------------------------------|
| FR-9.1  | Floating panel anchored to bottom-right of every page                          |
| FR-9.2  | States: closed (button only) → open (full 400px wide panel)                    |
| FR-9.3  | <b>Free and unlimited for ALL users</b> – no message caps, no login requi      |
| FR-9.4  | Keyboard shortcut: Ctrl+/ (Cmd+/) toggles panel                                |
| FR-9.5  | When on a topic page: topic title + course name injected into syst             |
| FR-9.6  | "Explain this topic" button → pre-fills chat with current topic ti             |
| FR-9.7  | "Ask AI about this code" → injects code into chat input                        |
| FR-9.8  | Suggested prompt chips shown when chat is empty                                |
| FR-9.9  | All requests proxied via <code>/api/ai/chat</code> – client never calls AI ser |
| FR-9.10 | Responses streamed token-by-token (ReadableStream / SSE)                       |
| FR-9.11 | Fallback: "AI assistant is temporarily unavailable" if service is              |
| FR-9.12 | Chat history persists in client state for the session (NOT stored              |

## 2.6 Admin Panel

| Module                         | Key Capabilities                                                                             |
|--------------------------------|----------------------------------------------------------------------------------------------|
| <b>Dashboard</b>               | Stats cards (users, MRR, DAU, content), revenue bar cha                                      |
| <b>Content – Block Editor</b>  | Create/edit Topics & Blogs. 14 block types. Drag-and-dr<br>save draft every 30s.             |
| <b>Content – Metadata</b>      | Slug (auto + editable), course/section assign, isPremi<br>tags, SEO fields                   |
| <b>Content – Image Upload</b>  | Option A: Cloudinary upload → URL. Option B: Static upl<br>Both in same dialog.              |
| <b>Navigation Builder</b>      | Header links (add/remove/reorder), dropdown menus, CTA<br>to <code>SiteConfig</code> DB doc. |
| <b>SEO Manager</b>             | Per-content meta title/description/OG image. Global SEC<br>Sitemap regenerate.               |
| <b>User Management</b>         | Search/filter/paginate users. Grant/revoke Pro. Change                                       |
| <b>Subscription Management</b> | View active/cancelled subs. Razorpay refund API. Webhoc                                      |



Block Type

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## 2.7 Subscription & Payments

| Plan        | Access                      | Price  |
|-------------|-----------------------------|--------|
| Free        | All free topics + all blogs | Free   |
| Pro Monthly | All topics + all blogs      | ₹299/r |
| Pro Annual  | All topics + all blogs      | ₹2,495 |

**No trial period.** Free content is always accessible. Premium requires payment.

## 2.8 Search

| ID      | Requirement                                                                   |
|---------|-------------------------------------------------------------------------------|
| FR-10.1 | Global Ctrl+K search modal – available on all pages                           |
| FR-10.2 | MongoDB text index on: topic title, blog title, course title, ta              |
| FR-10.3 | Results grouped by type: Topics / Blogs / Courses                             |
| FR-10.4 | Lock icon on gated topic results                                              |
| FR-10.5 | Recent searches stored in <code>localStorage</code> – shown as default sugges |
| FR-10.6 | Blog listing page has tag-based filter + text search                          |

## 2.9 Navigation Architecture

| Element              | Behaviour                                                        |
|----------------------|------------------------------------------------------------------|
| Header               | Sticky. Logo + nav links + search icon + theme togg              |
| Announcement bar     | Above header, configurable from admin (text, link, c             |
| "Learn" dropdown     | Hover → shows all courses with icons, DB-driven                  |
| User avatar dropdown | Dashboard, Billing, Logout                                       |
| Mobile               | Hamburger menu drawer                                            |
| Course sidebar       | Course title + progress bar + collapsible sections +             |
| Completed topic      | Green checkmark in sidebar                                       |
| Premium topic        | Lock icon + amber "PRO" badge for non-premium users              |
| Table of Contents    | Auto-generated from <code>heading</code> blocks. Highlights curr |
| Prev/Next buttons    | Bottom of each topic – navigates to adjacent topic i             |
| Breadcrumbs          | Home > Learn > [Course] > [Section] > [Topic]                    |
| Reading progress bar | Thin accent line at very top of page – fills as user             |

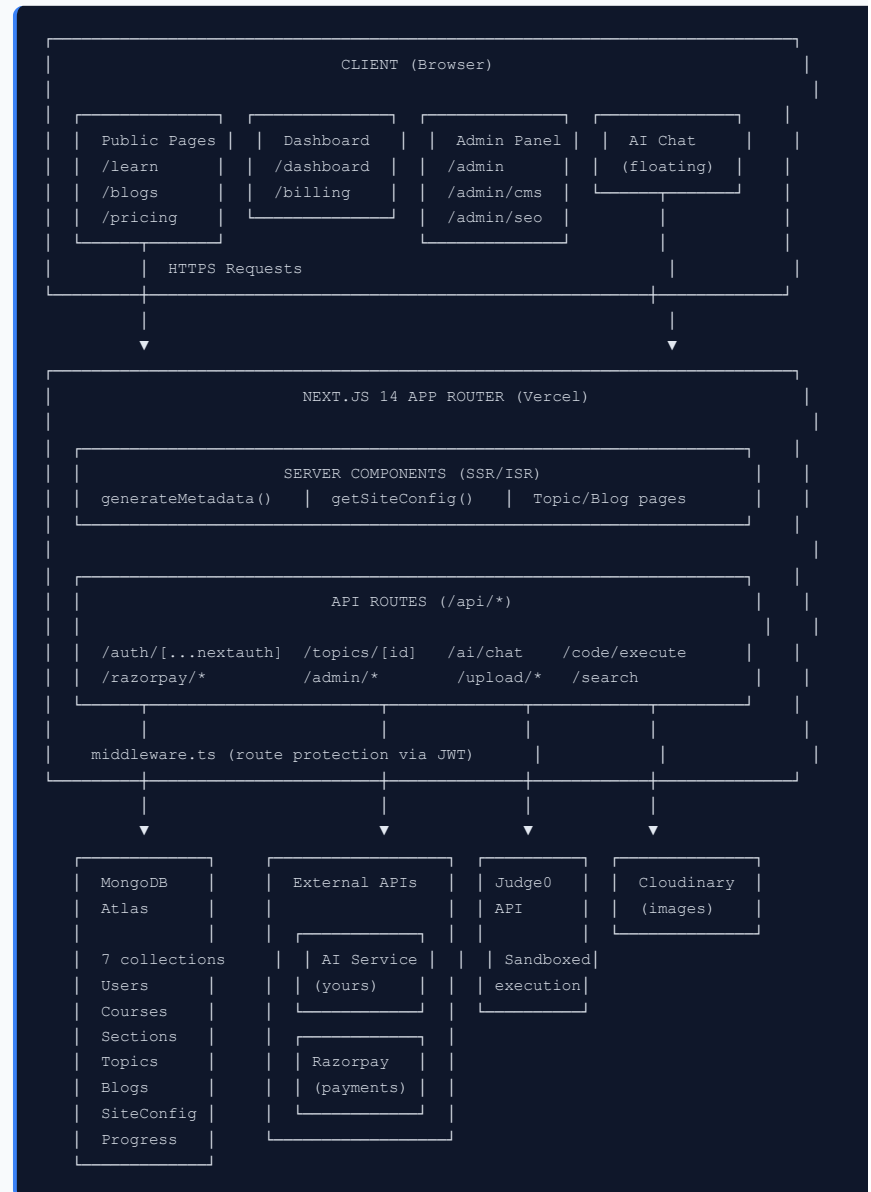
Block Type

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Shape

Rendered As

### 3. High-Level Architecture

#### 3.1 System Architecture Diagram



Block Type

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Shape

Rendered As

### 3.2 Request Lifecycle



## 4. Data Architecture – MongoDB Schemas

### 4.1 Entity Relationship Overview



Block Type

content

Shape

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## 4.2 Users Collection

```
{
  _id:           ObjectId,
  email:         String,      // unique, indexed
  name:          String,
  image:         String,      // avatar URL
  hashedPassword: String,     // null for OAuth users
  provider:      'email' | 'google' | 'github',

  // Role & Subscription
  role:          'free_user' | 'premium_user' | 'author' | 'admin' | 'super_admin',
  subscriptionStatus: 'free' | 'pro' | 'cancelled',
  razorpayPaymentId: String,    // last successful payment ID
  currentPeriodEnd:  Date,      // when Pro access expires

  // Profile
  bio:           String,
  github:        String,       // github.com/username
  linkedin:      String,       // linkedin.com/in/username

  // Activity
  streak:        Number,       // consecutive login days
  lastActiveDate: Date,
  theme:         'light' | 'dark' | 'system',
  emailNotifications: Boolean,

  createdAt:     Date,
  updatedAt:     Date,
}

Indexes:
  email: unique
```

## 4.3 Courses Collection

```
{
  _id:           ObjectId,
  slug:          String,      // unique, indexed - URL identifier
  title:         String,
  description:    String,
  icon:          String,      // emoji or image URL
  thumbnail:     String,      // cover image
  category:      'system-design' | 'l1d' | 'h1d' | 'kafka' | 'aws' | 'genai' | 'spark',
  difficulty:    'beginner' | 'intermediate' | 'advanced',
  tags:          [String],
  authorId:      ObjectId,    // ref: Users
  totalTopics:   Number,      // denormalized count (updated on topic create/delete)
  estimatedHours: Number,
  status:        'draft' | 'published' | 'archived',

  // SEO
  metaTitle:     String,
  metaDescription: String,
  ogImage:       String,

  publishedAt:   Date,
  createdAt:     Date,
  updatedAt:     Date,
}
```

Block Type

content

Shape

Rendered As

#### 4.4 Sections Collection

```
{
  _id:      ObjectId,
  courseId: ObjectId,    // ref: Courses, indexed
  slug:     String,
  title:    String,
  description: String,
  order:    Number,      // display order within course
  createdAt: Date,
}
```

#### 4.5 Topics Collection ← Core Content Document

```
{
  _id:      ObjectId,
  sectionId: ObjectId,    // ref: Sections, indexed
  courseId: ObjectId,    // ref: Courses, indexed (for queries without joining)
  slug:     String,      // indexed
  title:    String,      // nav label
  heading:  String,      // display H1 (can differ from nav title)
  order:    Number,      // position within section

  // THE CONTENT
  blocks: [
    {
      type:    String,      // see block types table in §2.2.2
      content: Mixed,       // structure varies by type
    }
  ],

  // Gating
  isPremium: Boolean,     // default: false
  previewBlockCount: Number, // default: 3 — blocks shown to free users

  // Metadata
  estimatedReadTime: Number, // minutes
  tags:              [String],
  status:            'draft' | 'published' | 'archived',
  scheduledAt:      Date,    // future publish time (optional)

  // SEO
  metaTitle:    String,
  metaDescription: String,
  ogImage:     String,

  publishedAt:  Date,
  createdAt:   Date,
  updatedAt:   Date,
}

Indexes:
{ courseId: 1, sectionId: 1, order: 1 }
{ slug: 1 }
Text index: { title: 'text', tags: 'text' }
```

Block Type

content  
Shape

Rendered As

## 4.6 Blogs Collection

```
{
  _id:      ObjectId,
  slug:     String,    // unique, indexed
  title:    String,
  subtitle: String,
  blocks:   [BlockSchema], // IDENTICAL format to Topics
  // NO isPremium field - all blogs are always free
  tags:     [String],
  coverImage: String,
  authorId: ObjectId,  // ref: Users
  rating:   Number,    // 0-5
  status:   'draft' | 'published' | 'archived',

  // SEO
  metaTitle:   String,
  metaDescription: String,
  ogImage:     String,

  publishedAt: Date,
  createdAt:  Date,
  updatedAt:  Date,
}

Text Index: { title: 'text', subtitle: 'text', tags: 'text' }
```

## 4.7 SiteConfig Collection (Key-Value Store)

```
{
  _id:      ObjectId,
  key:      String,    // unique: 'navigation' | 'global-seo'
  data:     Mixed,     // flexible JSON payload (see example below)
  updatedAt: Date,
  updatedBy: ObjectId, // ref: Users (which admin changed this)
}
```

navigation document data shape:

```
{
  "header": {
    "links": [
      { "label": "Learn", "type": "dropdown", "href": "/learn", "visible": true,
        "children": [
          { "label": "System Design", "href": "/learn/system-design", "icon": "📖" },
          { "label": "Apache Kafka", "href": "/learn/kafka", "icon": "⚡" }
        ]
      },
      { "label": "Blogs", "type": "link", "href": "/blogs", "visible": true },
      { "label": "Pricing", "type": "link", "href": "/pricing", "visible": true },
      { "label": "Roadmaps", "type": "link", "href": "/roadmaps", "visible": false }
    ],
    "ctaLabel": "Get Started",
    "ctaHref": "/register"
  },
  "announcement": {
    "enabled": true,
    "text": "🚀 New: AWS for Engineers - Use code LAUNCH50 for 50% off",
    "link": "/learn/aws",
    "bgColor": "#2563eb"
  },
  "footer": {
    "description": "Learn system design, Kafka, and more - built by engineers.",
    "links": [{ "label": "About", "href": "/about" }],
    "socials": [{ "platform": "github", "url": "https://github.com/..." }]
  }
}
```

Block Type

**content**  
Shape

Rendered As

#### 4.8 Progress Collection

```
{
  _id:           ObjectId,
  userId:        ObjectId,    // ref: Users
  courseId:      ObjectId,    // ref: Courses
  completedTopics: [ObjectId], // array of completed Topic IDs
  lastVisitedTopicId: ObjectId, // for "Resume Learning" button
  completionPercentage: Number, // 0-100

  updatedAt: Date,
}

Index: { userId: 1, courseId: 1 } - unique compound index
```

Block Type

content  
Shape

Rendered As

## 5. API Architecture

### 5.1 Complete API Route Map

```

app/api/
├── auth/
│   └── [...nextauth]/route.ts      POST - NextAuth handlers
├── courses/
│   ├── route.ts                  GET - list published courses
│   └── [courseId]/
│       ├── route.ts              GET - course metadata
│       └── sections/route.ts     GET - sections + topics list (no block conte
├── topics/
│   └── [topicId]/route.ts         GET - full topic (premium-gated server-side)
├── blogs/
│   ├── route.ts                  GET - blog list ?page=&category=&search=
│   └── [slug]/route.ts           GET - full blog (always public)
├── search/route.ts               GET - ?q=&type=all|topics|blogs|courses
├── user/
│   ├── me/route.ts               GET, PUT - profile (auth required)
│   ├── progress/route.ts         GET, POST - mark topic complete (auth require
│   └── bookmarks/
│       ├── route.ts              GET, POST - list, add bookmark
│       └── [id]/route.ts         DELETE - remove bookmark
├── ai/
│   └── chat/route.ts              POST - streaming proxy to AI service (open to
├── code/
│   └── execute/route.ts           POST - Judge0 proxy (rate limited 10/min/IP)
├── razorpay/
│   ├── create-order/route.ts     POST - create Razorpay order (auth required)
│   ├── verify-payment/route.ts   POST - HMAC verify + update subscription
│   └── webhook/route.ts          POST - async Razorpay events
├── upload/
│   ├── cloudinary/route.ts       POST - multipart → Cloudinary → returns { url
│   └── static/route.ts           POST - multipart → /public/uploads/ → { path
└── admin/
    (admin/author role required on all routes)
    ├── stats/route.ts            GET - dashboard stats
    ├── content/
    │   ├── topics/
    │   │   ├── route.ts          GET, POST - list all, create new
    │   │   └── [id]/route.ts     GET, PUT, DELETE - read, update, archive
    │   └── blogs/
    │       ├── route.ts          GET, POST
    │       └── [id]/route.ts     GET, PUT, DELETE
    ├── courses/route.ts          GET, POST, PUT - manage courses & sections
    ├── navigation/route.ts       GET, PUT - SiteConfig 'navigation' document
    ├── seo/
    │   └── [type]/[id]/route.ts   GET, PUT - SEO fields for topic|blog|course
    ├── users/
    │   ├── route.ts              GET - paginated user list
    │   └── [id]/route.ts         PUT - update role/subscription/ban
    ├── subscriptions/
    │   ├── route.ts              GET - all subscriptions
    │   └── refund/[id]/route.ts   POST - issue Razorpay refund

```



Block Type

**content**  
Shape

Rendered As

## 5.2 Response Shape Conventions

```
// Success
{ data: T, error: null }

// Error
{ data: null, error: { code: string, message: string } }

// Gated topic (partial content)
{
  ...topicFields,
  blocks: Block[],           // sliced to previewBlockCount
  isGated: true,
  totalBlocks: number,       // so client knows how much is hidden
}

// Full topic
{
  ...topicFields,
  blocks: Block[],           // all blocks
  isGated: false,
}
```

Block Type

content

Shape

Rendered As

## 6. Authentication & Authorization

### 6.1 Auth Flow Diagram

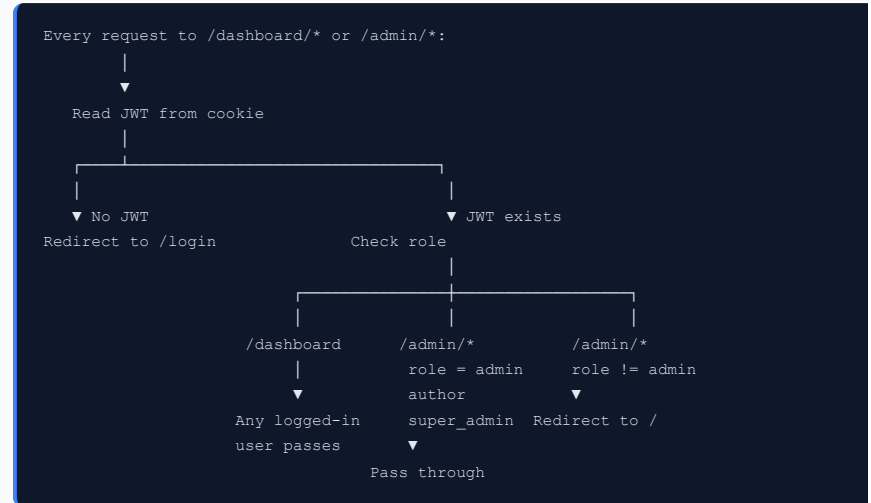


Block Type

content  
Shape

Rendered As

## 6.2 Route Protection – middleware.ts



## 6.3 Permission Levels in API Routes

```

// Three levels of protection:

// 1. Public – no auth check
export async function GET(req: Request) {
  // anyone can call this
}

// 2. Session required (dashboard routes)
export async function GET(req: Request) {
  const session = await auth();
  if (!session) return Response.json({ error: 'Unauthorized' }, { status: 401 });
}

// 3. Admin/Author only
export async function GET(req: Request) {
  const session = await auth();
  const adminRoles = ['admin', 'super_admin', 'author'];
  if (!session || !adminRoles.includes(session.user.role)) {
    return Response.json({ error: 'Forbidden' }, { status: 403 });
  }
}
  
```

## 7. Content System & Gating

## 7.1 Content Publishing Workflow



Block Type

content

Shape

Rendered As

## 7.2 Content Gating – Server-Side Enforcement

```

GET /api/topics/[topicId]
|
v
Topic.findById(topicId)
|
topic.isPremium?
|
├── NO
|   v
|   Return all
|   blocks[]
|   isGated: false
|
└── YES
    |
    | Check session JWT
    |
    | ┌── subscriptionStatus === 'pro'
    | |   AND currentPeriodEnd > new Date()
    | |   |
    | |   ├── YES
    | |   |   v
    | |   |   Return all
    | |   |   blocks[]
    | |   |   isGated: false
    | |   |
    | |   └── NO
    | |       v
    | |       Return
    | |       blocks.slice(0, previewBlockCount)
    | |       isGated: true
    | |       totalBlocks: topic.blocks.length
    | |
    | └── role is admin/author?
    |     |
    |     └── YES
    |         v
    |         Return all blocks[]
    |         isGated: false
  
```

## 7.3 Client-Side Gating UX (UX only – NOT security)

```

ContentRenderer component receives { blocks, isGated, totalBlocks }
|
isGated === true?
|
├── YES
|   v
|   Render preview blocks
|   +
|   Blur gradient overlay (CSS)
|   +
|   Lock icon + "Unlock Premium" CTA
|   |
|   User clicks "Unlock"
|   |
|   user logged in?
|   |
|   ├── NO
|   |   v
|   |   Show "Login to Continue" modal
|   |   |
|   |   └── Redirect to login with callbackUrl
|   |
|   └── YES
|       |
|       subscriptionStatus?
|       |
|       ├── free
|       |   v
|       |   Show "Subscribe to Unlock" modal
|       |   |
|       |   └── Razorpay flow (see $8)
|       |
|       └── pro (shouldn't happen)
|           v
|           Full access (reload page)
|
└── NO
    v
    Render all blocks (full content visible)
  
```

Block Type

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## 8. Payment System – Razorpay

### 8.1 Complete Payment Flow

```

User on /pricing or upgrade modal
|
User clicks "Upgrade to Pro"
|
▼
Step 1: Create Order


---


POST /api/razorpay/create-order
{ planType: 'monthly' | 'annual' }
|
Server: razorpay.orders.create({
  amount: 29900 | 249900,    // paise
  currency: 'INR',
  receipt: `r_${Date.now()}`
})
|
Returns: { orderId, amount, currency, key: RAZORPAY_KEY_ID }
|
▼
Step 2: Razorpay Checkout Modal (client-side)


---


const rzp = new Razorpay({
  key,
  amount,
  currency,
  order_id: orderId,
  name: 'EngineerTutorial Pro',
  prefill: { email: user.email, name: user.name },
  handler: async (response) => { ... } // called on success
})
rzp.open() // Razorpay UI modal opens in browser
|
User completes payment (UPI / card / netbanking)
|
Razorpay calls handler({ razorpay_order_id, razorpay_payment_id, razorpay_signature })
|
▼
Step 3: Verify Payment (server-side) – CRITICAL SECURITY STEP


---


POST /api/razorpay/verify-payment
{ razorpay_order_id, razorpay_payment_id, razorpay_signature }
|
Server computes:
  body = orderId + '|' + paymentId
  expected = HMAC-SHA256(body, RAZORPAY_KEY_SECRET)
|
expected === signature?
|
├── NO (tampered)
|   ▼
|   Return 400
|   "Invalid signature"
|   (do NOT grant access)
|
└── YES (valid)
    ▼
    Update MongoDB:
    User.subscriptionStatus = 'pro'
    User.currentPeriodEnd = now + 30 days
    User.razorpayPaymentId = paymentId
    |
    Return { success: true }
    |
    Client: window.location.reload()
    User now has premium access
    |
    ▼
Step 4: Webhook (async confirmation – belt-and-suspenders)


---


POST /api/razorpay/webhook
(called by Razorpay servers, not the client)
|

```

Block Type

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```

Verify X-Razorpay-Signature header
|
Handle event:
  payment.captured → ensure subscription is active
  payment.failed   → log, notify user
  subscription.charged → renew currentPeriodEnd

```

## 9. Code Execution System

### 9.1 Architecture



9.2 Rate Limiter Implementation

```
// Node.js in-memory Map - simple, sufficient for MVP
// Resets on server restart (Vercel functions spin down after inactivity)

const store = new Map<string, { count: number; resetAt: number }>();

export function checkRateLimit(key: string, limit: number, windowMs: number): boolean {
  const now = Date.now();
  const entry = store.get(key);

  if (!entry || now > entry.resetAt) {
    store.set(key, { count: 1, resetAt: now + windowMs });
    return true; // allowed - first request in window
  }

  if (entry.count >= limit) return false; // blocked
  entry.count++;
  return true; // allowed
}

// Key pattern: `code:${ip}` - 10 runs per minute per IP
// For authenticated users: could use `code:user:${userId}` instead
```

10. AI Chatbot Integration

10.1 Component Architecture





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## 10.2 AI Request &amp; Streaming Flow

```

User types message + hits Enter/Send
    |
    ▼
POST /api/ai/chat
{
  messages: [{ role: 'user', content: "What is load balancing?" }],
  contextTopic: "Load Balancing Deep Dive",
  contextCode?: "def my_func(): ..." // if "Ask AI" button was used
}
    |
    ▼
/api/ai/chat route:
  Build system prompt:
    "You are a CS tutor on EngineerTutorial.
    The user is reading: 'Load Balancing Deep Dive'.
    Answer concisely and clearly."
    |
  Prepend system message to messages array
  If contextCode: append code as user message
    |
  Forward to AI service:
  POST process.env.AI_SERVICE_URL
  Authorization: Bearer AI_SERVICE_API_KEY
  { messages: [...], stream: true }
    |
  Return streaming response directly to client:
  return new Response(aiResponse.body, {
    headers: { 'Content-Type': 'text/plain; charset=utf-8' }
  })
    |
    ▼
Client reads stream:
const reader = res.body.getReader()
while (true) {
  const { done, value } = await reader.read()
  if (done) break
  aiContent += decoder.decode(value)
  // Update last message in chat state in real-time
  updateLastMessage(aiContent)
}

```

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## 11. Admin Panel Architecture

### 11.1 Admin Panel Component Tree

```

AdminLayout
├── Sidebar
│   ├── Logo
│   ├── NavItem: Dashboard    → /admin
│   ├── NavItem: Content     → /admin/content
│   ├── NavItem: Navigation  → /admin/navigation
│   ├── NavItem: SEO         → /admin/seo
│   ├── NavItem: Users       → /admin/users
│   └── UserInfo + Logout
├── TopBar
│   ├── Page title (h1)
│   ├── Search input
│   ├── Notifications bell
│   └── "View Site" link
└── Page Content (children)
    ├── /admin                → DashboardPage
    ├── /admin/content        → ContentListPage
    ├── /admin/content/new    → BlockEditorPage (create)
    ├── /admin/content/[id]   → BlockEditorPage (edit)
    ├── /admin/navigation     → NavigationBuilderPage
    ├── /admin/seo            → SEOManagerPage
    └── /admin/users          → UserManagementPage
  
```

Block Type

content  
Shape

Rendered As

## 11.2 Block Editor Architecture

```

BlockEditorPage
|
|— State:
|   blocks: Block[]           (the content being edited)
|   metadata: TopicMetadata   (slug, isPremium, course, etc.)
|   status: 'draft'|'published'
|   isPreviewMode: boolean
|   lastSavedAt: Date
|
|— Auto-save: every 30s → PUT /api/admin/content/topics/[id]
|
|— Layout: 3-column
|   |
|   |— Column 1: Block Palette (left, 200px)
|   |   Clickable buttons for each block type:
|   |   [+ Heading] [+ Text] [+ Code] [+ Image]
|   |   [+ Callout] [+ Table] [+ Quiz] [+ Video]
|   |   [+ List] [+ Ordered List] [+ Badge] [+ Line]
|   |   → addBlock(type) appends new block to blocks[]
|   |
|   |— Column 2: Block List (center, flex-1)
|   |   DndContext (@dnd-kit/core)
|   |   SortableContext (blocks.map(b => b.id))
|   |   SortableBlock (for each block)
|   |   |— DragHandle (GripVertical icon)
|   |   |— BlockEditor (inline editing, varies by type)
|   |   |   |— HeadingEditor → <input>
|   |   |   |— TextEditor   → <textarea>
|   |   |   |— CodeEditor   → <MonacoEditor>
|   |   |   |— ImageEditor   → URL input + upload dialog
|   |   |   |— CalloutEditor → type selector + textarea
|   |   |   |— TableEditor   → dynamic rows/cols editor
|   |   |   |— QuizEditor    → question + options + answer
|   |   |   |— DeleteButton
|   |   |
|   |   |— Column 3: Metadata + Live Preview (right, 320px)
|   |   |   isPreviewMode = false → MetadataSidebar:
|   |   |   Slug (auto + editable)
|   |   |   Course assignment dropdown
|   |   |   Section assignment dropdown
|   |   |   Order number
|   |   |   isPremium toggle
|   |   |   Estimated read time
|   |   |   Tags (comma-separated or chips)
|   |   |   SEO: metaTitle, metaDescription, ogImage URL
|   |   |   Status: [Draft ▼] → [Published] → [Archived]
|   |   |   [Save Draft]   [Publish Now]
|   |   |
|   |   |   isPreviewMode = true → LivePreview:
|   |   |   <ContentRenderer blocks={blocks} />
|   |   |   (exact same component end users see)
|   |
|   |— Image Upload Dialog (modal):
|   |   Tab 1: Cloudinary Upload
|   |   |   [Choose file] → POST /api/upload/cloudinary → returns { url }
|   |   |   → insert url into image block content.src
|   |   Tab 2: Static Upload
|   |   |   [Choose file] → POST /api/upload/static → returns { path }
|   |   |   → insert path into image block content.src
|   |   Tab 3: URL
|   |   |   [Enter image URL] → insert directly

```

Block Type

content

Shape

Rendered As

### 11.3 Navigation Builder Data Flow

```

NavigationBuilderPage
|
▼
On page load: GET /api/admin/navigation
→ SiteConfig.findOne({ key: 'navigation' })
→ Populate form with current config
|
Admin edits:
- Toggle announcement bar on/off
- Change announcement text / link / color
- Add/remove/reorder header links
- Edit CTA button label and href
- Edit footer description
|
Live Preview Panel updates in real-time
(React state drives the preview, not the DB)
|
Admin clicks "Save Navigation"
|
▼
PUT /api/admin/navigation
{ header, announcement, footer }
|
SiteConfig.findOneAndUpdate({ key: 'navigation' }, { data: payload }, { upsert: true })
|
revalidatePath('/', 'layout') ← ALL pages using layout get fresh nav
|
Return { success: true }
|
Show "✓ Changes saved!" toast
|
Next user request to any page → sees updated navigation

```

## 12. SEO Architecture

### 12.1 Metadata per Route

```

// Per-route generateMetadata() — Next.js fetches metadata at build/request time

// Dynamic topic page
export async function generateMetadata({ params }): Promise<Metadata> {
  const topic = await Topic.findOne({ slug: params.topicId }).lean();
  return {
    title:      topic?.metaTitle || `${topic?.title} | EngineerTutorial`,
    description: topic?.metaDescription,
    openGraph: {
      title:      topic?.metaTitle || topic?.title,
      description: topic?.metaDescription,
      images: [{
        url: topic?.ogImage || `/og?title=${encodeURIComponent(topic?.title)}`,
        width: 1200,
        height: 630,
      }],
      type: 'article',
    },
    twitter: { card: 'summary_large_image' },
    alternates: {
      canonical: `/learn/${params.courseId}/${params.sectionId}/${params.topicId}`,
    },
  };
}

```

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## 12.2 SEO Architecture Diagram

```

| SEO LAYERS
|
| Layer 1: Per-page <head> metadata
| _____
| generateMetadata() in each page.tsx
| → <title>, <meta description>, <link canonical>
| → og:title, og:description, og:image, og:type
| → twitter:card, twitter:site
|
| Layer 2: Dynamic OG Images
| _____
| app/og/route.tsx
| GET /og?title=Load+Balancing
| → ImageResponse (1200×630 PNG)
| → Topic title + EngineerTutorial branding
| → Cached by Vercel CDN
|
| Layer 3: Structured Data (JSON-LD)
| _____
| Topic pages: Article + BreadcrumbList schema
| Course pages: Course schema
| Blog pages: Article schema
| FAQ sections: FAQPage schema
| → <script type="application/ld+json"> in <head>
|
| Layer 4: Sitemap
| _____
| app/sitemap.ts
| → Query MongoDB for all published topics, blogs, courses
| → Returns MetadataRoute.Sitemap[]
| → Served as /sitemap.xml
| → Regenerated via revalidatePath on admin publish
|
| Layer 5: robots.txt
| _____
| app/robots.ts
| Allow: all public pages
| Disallow: /admin/*, /api/*, /dashboard/*

```

## 12.3 On-Demand ISR Flow

```

Admin publishes new Topic
|
| ▼
PUT /api/admin/content/topics/[id]
{ status: 'published', blocks: [...] }
|
MongoDB updated
|
revalidatePath(`/learn/${courseSlug}/${sectionSlug}/${topicSlug}`)
revalidatePath('/sitemap.xml')
|
Next.js ISR cache for those paths → INVALIDATED
|
Next user request to that topic URL:
→ Full SSR (topic fetched from DB, fresh)
→ Response cached again for subsequent requests
|
All other topic pages: still served from cache (fast)
Only the changed page gets fresh SSR

```

Block Type

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Shape

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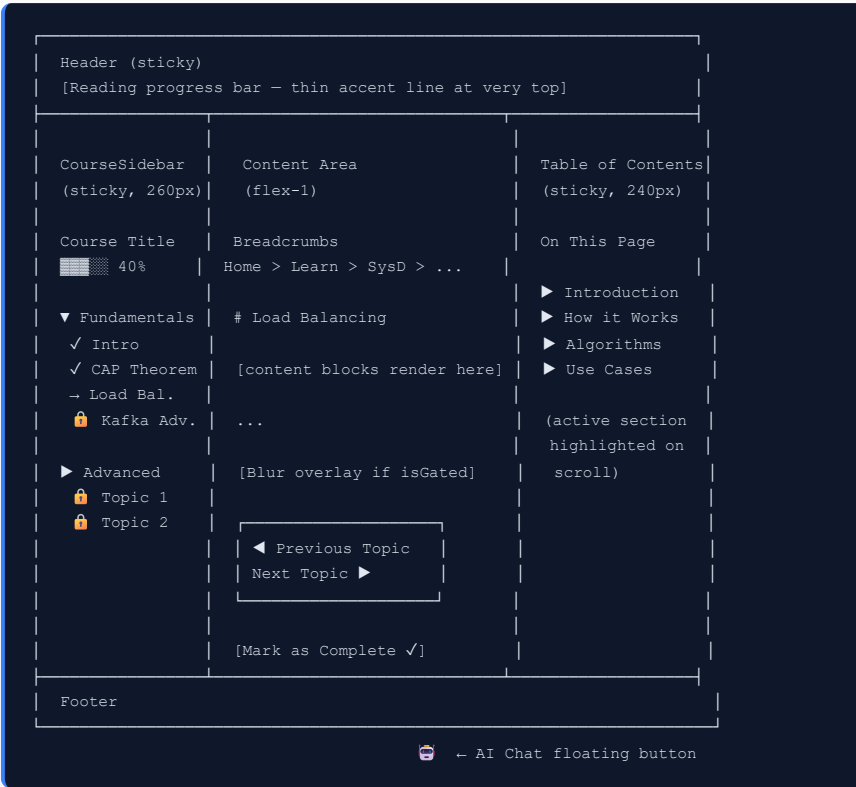
## 13. Frontend Architecture

### 13.1 Page Structure

```

app/
├── (public)/
│   ├── page.tsx                No auth required
│   │                               Landing page
│   ├── blogs/
│   │   ├── page.tsx            Blog listing – tag filters, search, pagination
│   │   └── [slug]/page.tsx      Blog detail – full content, related posts, AI c
│   ├── learn/
│   │   ├── page.tsx            All courses list
│   │   └── [courseId]/
│   │       ├── page.tsx        Course overview – sections preview, start butto
│   │       └── [sectionId]/[topicId]/
│   │           └── page.tsx     Topic reader – 3-panel layout
│   ├── playground/page.tsx     Full-screen code editor
│   └── pricing/page.tsx         Plan comparison, FAQ, Razorpay CTA
├── (auth)/
│   ├── login/page.tsx          Split layout – social + email form
│   └── register/page.tsx       Split layout – name/email/password + strength me
├── (dashboard)/
│   └── dashboard/
│       ├── page.tsx            User home – stats, in-progress, explore
│       ├── bookmarks/page.tsx
│       └── billing/page.tsx
└── (admin)/
    └── admin/
        ├── page.tsx            Admin dashboard
        ├── content/
        │   ├── page.tsx        Content table
        │   └── new/page.tsx     Block editor (create)
        ├── navigation/page.tsx
        ├── seo/page.tsx
        └── users/page.tsx
  
```

13.2 Topic Reader — 3-Panel Layout



13.3 State Management Architecture

```
Global State (Context API)
├─ AuthContext (lib/auth.tsx)
│  └─ user: User | null
│  └─ isLoading: boolean ← CRITICAL: prevents redirect before localStorage loads
│  └─ login(email, password, name?) → void
│  └─ logout() → void
│  └─ isAdmin: boolean
│  └─ isPremium: boolean
├─ ThemeContext (context/ThemeContext.tsx)
│  └─ theme: 'light' | 'dark'
│  └─ toggleTheme() → void
│  └─ [persisted to localStorage + DB (user.theme)]
└─ ChatContext (context/ChatContext.tsx)
   └─ isOpen: boolean
   └─ messages: Message[]
   └─ isLoading: boolean
   └─ toggle() → void
   └─ addMessage(msg) → void
   └─ clearHistory() → void

Server State (TanStack React Query)
├─ useQuery(['courses']) → GET /api/courses
├─ useQuery(['topic', id]) → GET /api/topics/[id]
├─ useQuery(['progress']) → GET /api/user/progress
├─ useMutation → POST /api/user/progress (mark complete)
└─ useQuery(['bookmarks']) → GET /api/user/bookmarks
```

Block Type

content  
Shape

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## 14. Security Architecture

### 14.1 Security Layers

```

Layer 1: Transport Security
  → HTTPS everywhere (Vercel enforces TLS)
  → HSTS headers

Layer 2: Authentication
  → NextAuth.js JWT stored in httpOnly cookie (XSS-safe)
  → JWT contains: userId, role, subscriptionStatus, currentPeriodEnd
  → 30-day expiry with sliding window

Layer 3: Authorization
  → middleware.ts blocks /admin/* and /dashboard/* at edge
  → API routes re-check session with auth() (defense in depth)
  → Role checked both in middleware AND in each API route

Layer 4: Content Gating (MOST CRITICAL)
  → Server returns ONLY preview blocks for non-premium users
  → Client NEVER receives full content unless authorized
  → Client-side blur is UX only — removing it via DevTools
    reveals no additional content

Layer 5: Payment Integrity
  → Razorpay signature verified server-side (HMAC-SHA256)
  → Client NEVER receives RAZORPAY_KEY_SECRET
  → Subscription update only happens after signature verification

Layer 6: Input Validation
  → All API inputs validated with Zod schemas
  → File uploads: MIME type check + size limit (5MB)
  → Admin-only routes check role on every request

Layer 7: Rate Limiting
  → Code execution: 10 runs/minute/IP (in-memory)
  → Can add API-level rate limiting per route later

Layer 8: Injection Prevention
  → MongoDB queries use Mongoose (parameterized, no string interpolation)
  → Content blocks rendered as React elements (not dangerouslySetInnerHTML)
  → Exception: only trusted admin-authored content may use HTML; sanitize with DOMPurify

```

### 14.2 Environment Variable Security

```

Server-only (never sent to client):
MONGODB_URI      — DB connection string
NEXTAUTH_SECRET   — JWT signing secret
RAZORPAY_KEY_SECRET — payment signature key
GOOGLE_CLIENT_SECRET — OAuth secret
GITHUB_CLIENT_SECRET — OAuth secret
JUDGE0_API_KEY    — code execution API key
AI_SERVICE_API_KEY — AI service authentication
CLOUDINARY_API_SECRET — image upload secret

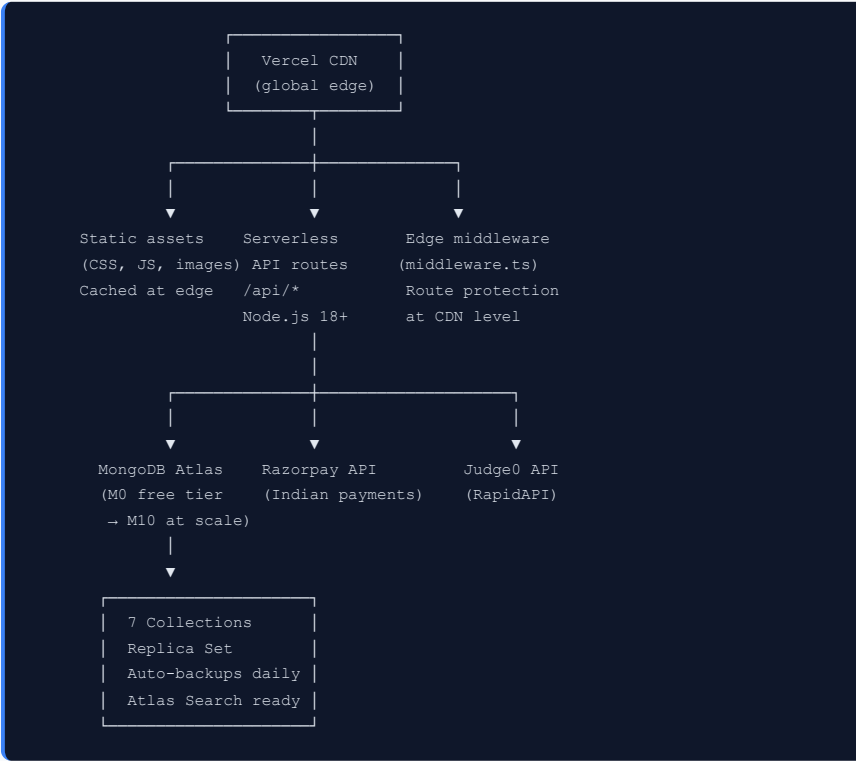
Client-safe (NEXT_PUBLIC_ prefix):
NEXT_PUBLIC_APP_URL — used for canonical URLs and OG images
NEXT_PUBLIC_RAZORPAY_KEY_ID — the public key (safe to expose)

```



15. Deployment Architecture

15.1 Infrastructure Overview



15.2 Environment Stages

| Stage       | URL                   | Branch      | DB          |
|-------------|-----------------------|-------------|-------------|
| Development | http://localhost:3000 | any         | Local Mongo |
| Preview     | *.vercel.app          | PR branches | Atlas Dev c |
| Production  | yourdomain.com        | main        | Atlas Produ |

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### 15.3 Deployment Checklist

#### Phase 1: MongoDB Atlas

- ☐ Create Atlas account → New Project → Build M0 Cluster (Free)
- ☐ Create DB user: username/password with readWrite on engineer-tutorial DB
- ☐ Network Access → Allow from Anywhere (0.0.0.0/0) – required for Vercel
- ☐ Get connection string → replace in MONGODB\_URI env var
- ☐ Enable Atlas Search index on Topics (title, tags) for full-text search

#### Phase 2: OAuth Setup

- ☐ Google: console.cloud.google.com → Credentials → OAuth 2.0  
Authorized redirect URI: https://yourdomain.com/api/auth/callback/google
- ☐ GitHub: github.com/settings/apps → New OAuth App  
Callback URL: https://yourdomain.com/api/auth/callback/github
- ☐ Generate NEXTAUTH\_SECRET: openssl rand -base64 32

#### Phase 3: Razorpay

- ☐ razorpay.com → Settings → API Keys → Generate Test Keys
- ☐ RAZORPAY\_KEY\_ID + RAZORPAY\_KEY\_SECRET → env vars
- ☐ Webhooks → Add webhook → URL: https://yourdomain.com/api/razorpay/webhook
- ☐ Enable events: payment.captured, payment.failed
- ☐ Before launch: switch to Live Keys

#### Phase 4: Judge0 (Code Execution)

- ☐ rapidapi.com → Judge0 CE → Subscribe (Free: 50 req/day)
- ☐ JUDGE0\_API\_KEY → env var

#### Phase 5: Cloudinary (Image Upload)

- ☐ cloudinary.com → Dashboard → API Keys
- ☐ CLOUDINARY\_CLOUD\_NAME + CLOUDINARY\_API\_KEY + CLOUDINARY\_API\_SECRET → env vars

#### Phase 6: Vercel Deploy

- ☐ vercel.com → Import GitHub repo
- ☐ Framework: Next.js (auto-detected)
- ☐ Add all env vars from .env.local
- ☐ Deploy → Production URL assigned
- ☐ Set NEXTAUTH\_URL = https://yourdomain.com
- ☐ Set NEXT\_PUBLIC\_APP\_URL = https://yourdomain.com

#### Phase 7: Content Migration

- ☐ Run: node scripts/migrate.mjs
- ☐ Seeds MongoDB from existing JSON files in reference repo
- ☐ Blocks[] copied directly – format is identical

#### Phase 8: Go Live

- ☐ Run Lighthouse audit on 3 pages (/ , /blogs/[slug], /learn/[c]/[s]/[t])
- ☐ Target: Performance > 90, SEO = 100, Accessibility > 90
- ☐ Switch Razorpay to Live Keys
- ☐ Submit sitemap.xml to Google Search Console
- ☐ Submit sitemap.xml to Bing Webmaster Tools

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#### 15.4 NPM Packages — Complete Install Command

```
# Core
npm install next@14 react react-dom typescript

# Database & Auth
npm install mongoose next-auth@beta bcryptjs

# Payments
npm install razorpay

# Code Editor
npm install @monaco-editor/react

# Block Editor DnD
npm install @dnd-kit/core @dnd-kit/sortable @dnd-kit/utilities

# Server State
npm install @tanstack/react-query

# Image Upload
npm install cloudinary

# Validation
npm install zod

# Image Optimization
npm install sharp

# Dev Dependencies
npm install -D tailwindcss postcss autoprefixer
npm install -D @types/node @types/react @types/react-dom @types/bcryptjs
npm install -D typescript eslint eslint-config-next
```

Block Type

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Shape

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## Appendix A — Block Content Schema Examples

```
// heading block
{ "type": "heading", "content": { "text": "What is Load Balancing?" } }

// text block
{ "type": "text", "content": { "text": "Load balancing distributes incoming traffi

// code block
{
  "type": "code",
  "content": {
    "language": "python",
    "code": "def round_robin(servers, request):\n    return servers[request.id % len(s
    "filename": "load_balancer.py"
  }
}

// callout block
{ "type": "callout", "content": { "type": "tip", "text": "Round-robin is simple but do

// quiz block
{
  "type": "quiz",
  "content": {
    "question": "Which algorithm sends requests to the server with fewest active conne
    "options": ["Round Robin", "Least Connections", "IP Hash", "Random"],
    "answer": 1
  }
}

// table block
{
  "type": "table",
  "content": {
    "headers": ["Algorithm", "Best For", "Drawback"],
    "rows": [
      ["Round Robin", "Equal servers", "Ignores load"],
      ["Least Connections", "Long-lived requests", "More overhead"]
    ]
  }
}
```

Block Type

content

Shape

Rendered As

### Appendix B — Key Files Reference

| File                                     | Purpose                        |
|------------------------------------------|--------------------------------|
| app/api/topics/[topicId]/route.ts        | Content gating — most security |
| lib/auth.tsx                             | Auth context + fake-auth :     |
| lib/auth.ts                              | NextAuth configuration (p:     |
| middleware.ts                            | Route protection at edge       |
| models/Topic.ts                          | Core content schema with l     |
| models/SiteConfig.ts                     | Navigation + SEO config s      |
| components/editor/BlockEditor.tsx        | Admin CMS — most complex c     |
| components/ai/AIChatWindow.tsx           | AI floating panel + strear     |
| app/api/razorpay/verify-payment/route.ts | HMAC signature verificatio     |
| app/api/code/execute/route.ts            | Judge0 proxy + rate limite     |
| app/sitemap.ts                           | Auto-generated sitemap fr      |
| scripts/migrate.mjs                      | One-time data migration f:     |

### 16. Future Features Roadmap

These features are **NOT** part of the MVP. Build them after the platform is 1 feedback. Each phase has a trigger condition — don't build it speculatively

#### 16.1 Phase Trigger Conditions

| Phase                  | Build When...                                      |
|------------------------|----------------------------------------------------|
| Phase 2 — Community    | DAU > 200 and users ask for comments/notes in feed |
| Phase 3 — Gamification | Week-2 retention < 40%. Gamification directly targ |
| Phase 4 — Video        | Monthly revenue > ₹2L AND users request video cont |
| Phase 5 — AI Expansion | Your AI service handles complex multi-turn convers |
| Phase 6 — Monetization | First B2B inbound inquiry. Don't build corporate f |
| Algolia Search         | Search latency > 800ms or relevance complaints fr  |
| Mobile App             | Mobile traffic > 50% AND PWA engagement scores are |

#### 16.2 Phase 2 — Community & Social Features (Months 4–6)

##### 2A. Per-Topic Comment Threads

**Why:** GFG and LeetCode see huge engagement through discussion sections. Users learn from e errors in content.

**Data Model:**

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|------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            |                  | <pre>// New collection: Comments {   _id:      ObjectId,   topicId:  ObjectId, // ref: Topics   userId:   ObjectId, // ref: Users   content:  String,    // markdown text   parentId: ObjectId,  // null for top-level, ref: Comments for replies   upvotes:  Number,   isDeleted: Boolean,   createdAt: Date, }</pre> <p><b>What to build:</b></p> <ul style="list-style-type: none"> <li>• Comment thread UI below every topic (collapsible)</li> <li>• Upvote / downvote buttons (stored per-user to prevent double voting)</li> <li>• One level of reply threading ( <code>parentId</code> pattern)</li> <li>• <code>[Author]</code> badge on replies by the topic's author</li> <li>• Report / flag button → admin can delete from admin panel</li> <li>• Comment count shown on topic card in sidebar</li> </ul> <p><b>API Routes:</b></p> <pre>GET  /api/topics/[id]/comments    → paginated, nested POST /api/topics/[id]/comments    → create (auth required) PUT  /api/comments/[id]/upvote    → toggle upvote DELETE /api/admin/comments/[id]    → admin delete</pre> <hr/> <p><b>2B. User Notes &amp; Highlights</b></p> <p><b>Why:</b> Users want to annotate content while learning – dramatically increases time-on-site</p> <p><b>Data Model:</b></p> <pre>// New collection: UserNotes {   userId:      ObjectId,   topicId:     ObjectId,   content:     String, // markdown notes text   highlights:  [{ text: String, color: String, startOffset: Number }],   updatedAt:   Date, }</pre> <p><b>What to build:</b></p> <ul style="list-style-type: none"> <li>• Private sticky notes textarea per topic (auto-saved to DB)</li> <li>• Text highlight: select content → color picker appears → saved with position offsets</li> <li>• "My Notes" page in user dashboard – lists all notes across all topics</li> <li>• Export notes as PDF (using <code>jsPDF</code> or Puppeteer server-side)</li> </ul> <hr/> <p><b>2C. Public Profile Pages</b></p> <p><b>URL:</b> <code>/u/[username]</code></p> <p><b>What to build:</b></p> <ul style="list-style-type: none"> <li>• Avatar, name, bio, GitHub/LinkedIn links</li> <li>• Activity summary: courses completed, topics completed, current streak</li> <li>• Badges earned (Phase 3 gamification unlocks this)</li> <li>• Optional: "Follow" button + activity feed ("John completed System Design Fundamentals")</li> <li>• Privacy toggle: user can make profile private</li> </ul> |

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### 2D. Notification System

**In-App Notifications:**

```
// New collection: Notifications
{
  userId:   ObjectId,
  type:     'comment_reply' | 'new_content' | 'streak_reminder' | 'sub_expiry',
  message:  String,
  link:     String,           // navigate to on click
  isRead:   Boolean,
  createdAt: Date,
}
```

**Email Notifications** (use Resend or Nodemailer):

- Weekly digest: "3 new topics published in Apache Kafka this week"
- Streak reminder: "You haven't visited in 2 days – keep your streak alive!"
- Subscription expiry: "Your Pro plan expires in 7 days"

**User Controls:** Per-type toggle in profile settings. User can opt out of each type individ

## 16.3 Phase 3 – Gamification & Certificates (Months 7-9)

### 3A. XP System & Levels

**Why:** Gamification (Duolingo, LeetCode, Codecademy) increases DAU and reduces churn. Users unlock badges.

**XP Events:**

| Action           | XP Gained    |
|------------------|--------------|
| Complete a topic | +10 XP       |
| Daily login      | +5 XP        |
| Comment upvoted  | +2 XP        |
| 7-day streak     | +50 XP bonus |
| First topic ever | +20 XP       |

**Level System:**

| Level | XP Range  | Title        |
|-------|-----------|--------------|
| 1     | 0-99      | Novice       |
| 2     | 100-499   | Learner      |
| 3     | 500-1499  | Practitioner |
| 4     | 1500-3999 | Expert       |
| 5     | 4000+     | Master       |

**DB Changes:** Add `totalXP: Number` to Users schema.

### 3B. Badges

**Badge Definitions** (awarded automatically on trigger):

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| Badge                  | Trigger                           |
|------------------------|-----------------------------------|
| "First Step"           | Complete first topic              |
| "System Design Expert" | Complete all System Design topics |
| "Kafka Master"         | Complete all Kafka topics         |
| "7-Day Streak"         | Login 7 consecutive days          |
| "Blog Explorer"        | Read 10 blog posts                |
| "Pro Member"           | Subscribe to Pro plan             |
| "Helpful"              | Receive 10 upvotes on comments    |

**DB Changes:** Add `badges: [{ badgeId: String, earnedAt: Date }]` to Users schema.

### 3C. Weekly Leaderboard

```
// Query: top 10 users by XP earned in last 7 days
// Store weekly XP separately to enable weekly reset
// New collection: WeeklyXP
{
  userId: ObjectId,
  weekOf: Date,      // Monday of the week
  xpEarned: Number,
}
```

- Weekly top 10 by XP earned in last 7 days
- Filter by course category (e.g., top Kafka learners this week)
- Resets every Monday at 00:00 IST

### 3D. Problem of the Day (POTD)

**Why:** Daily challenges create a habit loop that brings users back every single day.

```
// New collection: DailyProblems
{
  date: Date,        // unique per day
  title: String,
  difficulty: 'easy' | 'medium' | 'hard',
  description: String,
  starterCode: { [language: string]: string },
  testCases: [{ input: String, expectedOutput: String }],
  hints: [String],
  solution: String,
}
```

- POTD page shows today's problem; solved via code editor
- Solving awards +25 XP + streak update
- Past problems archived and accessible
- Badge: "30 POTDs Solved"
- Email reminder daily at 9 AM for opted-in users

### 3E. Course Completion Certificates

**Why:** Certificates give learners something tangible to show employers – major motivation to complete course

**Certificate Flow:**



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```
completionPercentage reaches 100%
|
Trigger certificate generation
|
Server-side: render HTML template → convert to PDF
(Puppeteer or jsPDF)
|
Certificate stored in MongoDB:
{ _id, userId, courseId, issuedAt, certificateId (UUID) }
|
Public URL: /certificates/[certificateId]
|
"Share on LinkedIn" deep link (pre-fills certification form)
"Download PDF" button
```

## 16.4 Phase 4 – Video & Live Content (Months 10–14)

### 4A. Video Courses

**Video block** ( `type: 'video'` ) already exists in the schema. Extend the implementation:

| Source            | Implementation                                            |
|-------------------|-----------------------------------------------------------|
| YouTube URL       | Auto-embed using <code>youtube-nocookie.com</code> iframe |
| Cloudinary Upload | Upload video → get URL → custom                           |
| Mux (scale)       | Mux for adaptive bitrate streaming at high traf           |

**Custom Video Player** ( `components/content/VideoPlayer.tsx` ):

- Built on HTML5 or Plyr.io
- Chapters, playback speed (0.75×, 1×, 1.25×, 1.5×, 2×)
- Keyboard shortcuts (space = play/pause, arrow = seek 10s)
- Auto-saves watch position (resume from where you left off)

**Video Transcript:**

- Auto-generated via Cloudinary transcription or AssemblyAI
- Displayed as scrollable text below video – syncs with playback
- Transcript text indexed for search (SEO goldmine)

### 4B. Live Coding Sessions

- Session calendar page – upcoming live events
- "RSVP" button – `.ics` download or Google Calendar link
- Live session via embedded `Daily.co` or `100ms.live` iframe
- In-session Q&A via platform chat
- Recordings auto-stored in Cloudinary → published as premium content post-session

### 4C. Interactive Roadmaps

**Why:** Roadmaps are high-traffic content (roadmap.sh gets millions of visits). They rank well for many queries.

**Tech:** `reactflow` library for node-graph rendering.

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- Visual node-graph per learning track (System Design, Kafka, etc.)
- Each node links to the relevant topic
- Completed topics shown with filled/colored nodes (synced with user progress)
- "Share my progress" → static image with completed nodes highlighted
- Each roadmap page targets "[Topic] learning roadmap" keywords for SEO

## 16.5 Phase 5 — AI Expansion (Months 12–18)

These features assume your separate AI service is mature and handles compl

### 5A. AI Mock Interviews

**Why:** HelloInterview's top differentiator. This is a massive moat once built well.

**Interview Types:**

| Type          | AI Role                                                            |
|---------------|--------------------------------------------------------------------|
| System Design | Presents design problem, asks follow-ups on scalability/trade-offs |
| LLD           | Presents class design problem                                      |
| Behavioral    | Asks STAR-format questions                                         |

**Scoring Report** (generated after each session):

```
// New collection: MockInterviews
{
  userId:      ObjectId,
  type:        'system-design' | 'lld' | 'behavioral',
  question:    String,
  transcript:  [{ role: String, content: String }],
  score: {
    architecture: Number,    // 0-10
    scalability:  Number,    // 0-10
    tradeoffs:    Number,    // 0-10
    overall:      Number,
  },
  feedback: {
    strengths: [String],
    missed:    [String],
  },
  createdAt: Date,
}
```

**UX:** Dedicated `/practice/[type]` page — full-page chat interface.

### 5B. AI Code Review

**Why:** Instant detailed code feedback at scale — impossible with human reviewers.

**Flow:**

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|------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            |                  | <pre> User runs code in Playground   "Get AI Review" button appears in output panel   POST /api/ai/review { code, language, output, topic? }   AI returns structured JSON: {   issues: [{ line: number, type: 'error' 'style' 'performance', message, suggestion }]   complexity: { time: 'O(n^2)', space: 'O(1)' },   overall: "Good approach but nested loops can be optimized..." }   Monaco Editor shows inline decorations (red/yellow underlines) Hover tooltip shows message + suggestion Side panel: diff view "Your code" vs "Suggested improvement" </pre>     |
|            |                  | <h3>5C. Personalized Learning Path</h3> <p><b>Skills Assessment Quiz</b> (shown on first login):</p> <ul style="list-style-type: none"> <li>• 10-15 MCQs covering DSA, System Design, Kafka basics</li> <li>• Based on score → AI recommends personalized course order</li> <li>• "Daily Recommendation" widget on dashboard: "Based on your progress, study [topic] next"</li> <li>• "You're weak in [area]" suggestions based on quiz + incomplete topics</li> <li>• Store in DB: <code>Users.assessmentScore: { category: string, score: number }[]</code></li> </ul> |
|            |                  | <h3>5D. AI Resume Review</h3> <p><b>Target audience:</b> Engineers preparing for interviews – high need for resume feedback.</p> <p><b>Flow:</b></p> <ul style="list-style-type: none"> <li>• PDF upload in user dashboard</li> <li>• AI extracts projects, skills, technologies, years of experience</li> <li>• Feedback: quantified impact suggestions, keyword optimization, formatting tips</li> <li>• "Suggested topics to study" based on gaps vs. target role requirements</li> <li>• Separate API call with resume-specific system prompt</li> </ul>             |
|            |                  | <h3>5E. AI-Generated Practice Problems</h3> <ul style="list-style-type: none"> <li>• "Generate Practice Problem" button on each topic page</li> <li>• AI generates a problem relevant to the current topic</li> <li>• Problem types: System Design question, LLD class design, MCQ, short answer</li> <li>• User answers evaluated by AI against a rubric</li> <li>• Strong answers stored as community problems (after admin review)</li> </ul>                                                                                                                         |
|            |                  | <h2>16.6 Phase 6 – Monetization Expansion (Months 18–24)</h2> <h3>6A. Corporate / Team Accounts</h3> <p><b>Why:</b> B2B revenue is 5–10× more valuable per user than B2C.</p>                                                                                                                                                                                                                                                                                                                                                                                            |

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|------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            |                  | <pre>// New collection: Teams {   _id:           ObjectId,   name:          String,   adminId:       ObjectId,    // ref: Users   members:       [ObjectId],  // ref: Users   assignedCourses: [ObjectId], // ref: Courses   planType:      'team',   seatCount:     Number,   invoiceEmail:  String,   createdAt:     Date, }</pre> <p><b>Features:</b></p> <ul style="list-style-type: none"> <li>• Team admin dashboard: member progress, assign courses, set completion deadlines</li> <li>• Per-seat pricing: ₹299/seat/month (minimum 5 seats)</li> <li>• Invoice-based billing (B2B companies need invoices, not card charges)</li> <li>• Private courses visible only to specific teams</li> <li>• Progress reports exportable as PDF for managers</li> </ul> <hr/> <p><b>6B. 1:1 Mentorship Marketplace</b></p> <p><b>Why:</b> HelloInterview charges \$90-\$120 per coaching session. Platform audience of motivated</p> <pre>// New collections: Mentors, Sessions, MentorReviews  // Mentors {   userId:       ObjectId,    // ref: Users (mentor's user account)   linkedIn:     String,   company:      String,   yearsExp:     Number,   specializations: [String], // e.g., ['system-design', 'kafka']   hourlyRate:   Number,     // INR   rating:       Number,   isVerified:   Boolean,    // admin-approved }  // Sessions {   mentorId:     ObjectId,   studentId:    ObjectId,   scheduledAt:  Date,   durationMin:  Number,   status:       'upcoming'   'completed'   'cancelled',   meetingUrl:   String,     // Daily.co or Zoom link   razorpayOrderId: String, }</pre> <p><b>Revenue split:</b> Platform 20%, mentor 80% — handled via Razorpay Transfer API.</p> <hr/> <p><b>6C. Job Board</b></p> <p><b>Why:</b> Target audience is actively job-seeking — creates another revenue stream and increas</p> |

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```
// New collection: JobListings
{
  title:      String,
  company:    String,
  location:   String,
  type:       'full-time' | 'contract' | 'remote',
  description: String,
  requiredSkills: [String],
  applyUrl:   String,
  postedBy:   ObjectId,    // company admin user
  expiresAt:  Date,
  status:     'active' | 'expired',
}
```

**Features:**

- "Apply with EngineerTutorial Profile" – sends completed courses + certificates
- "Verified Skills" badge on applications for users who completed relevant courses
- Job alerts: user sets preferences → email alerts for matching jobs
- Employer dashboard: post jobs, view applicants, filter by completed courses
- Pricing: ₹X per 30-day listing (revenue from job posts)

---

### 6D. Company-Specific Interview Prep Packs

**Why:** HelloInterview monetizes this effectively. Users pay premium for targeted prep.

- Dedicated packs (one-time purchase, NOT subscription): "Crack Google SWE L4", "Amazon
- Each pack: curated question bank with company tags, insider tips, past experiences (an company-specific scoring rubric
- Priced as one-time purchase (₹X) or bundled with Pro subscription
- **Packs** collection – separate from courses, company-specific content

---

### 6E. Content Licensing API

- REST API + LTI (Learning Tools Interoperability) for Canvas/Moodle integration
- Partner institutions embed content in their own LMS with an API key
- White-label option: **learn.university.edu** powered by EngineerTutorial, custom brandin
- Pricing: monthly licensing fee per institution based on student count
- Analytics API: partners see which content their students engage with

---

## 16.7 Ongoing Technical Improvements

### Search – Upgrade to Algolia or Meilisearch

**When:** MongoDB text search latency > 800ms or poor relevance from real users.

| Option      | Hosted      | Cost               | Best For          |
|-------------|-------------|--------------------|-------------------|
| Algolia     | Yes         | Free tier generous | Best relevance, f |
| Meilisearch | Self-hosted | Cheaper at scale   | Open source, sim: |

Both support: instant-as-you-type, highlighted matched terms, filter by **isPremium** / **cate**

---

### Analytics – PostHog

**When:** Immediately after launch.

- **Funnel tracking:** Landing → Signup → First Topic → Upgrade (find where users drop off)
- **Session recordings:** Watch real user sessions to spot UX issues
- **Feature flags:** A/B test pricing page CTA, topic layout, announcement bar text

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|-------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----------|-----------|-------------|------------|------------|----------|----------------------|------------------|-------------|------------------|--------------|
|             |                      | <ul style="list-style-type: none"> <li>• <b>Heatmaps:</b> Where users click, how far they scroll on topic pages</li> </ul> <p>Install: <code>npm install posthog-js</code> – initialize in <code>app/layout.tsx</code></p> <hr/> <p><b>A/B Testing – Pricing Page</b></p> <p><b>When:</b> 1,000+ monthly visitors to <code>/pricing</code>.</p> <p>Tests to run:</p> <table> <tr> <th>Test</th><th>Variant A</th><th>Variant B</th></tr> <tr> <td>Price point</td><td>₹199/month</td><td>₹299/month</td></tr> <tr> <td>CTA copy</td><td>"Unlock All Content"</td><td>"Start Learning"</td></tr> <tr> <td>Page layout</td><td>Comparison table</td><td>Feature list</td></tr> </table> <p>Use PostHog feature flags or GrowthBook. Track conversion rate per variant.</p> <hr/> <p><b>Internationalization (i18n)</b></p> <p><b>When:</b> Significant traffic from Hindi-speaking users (highly likely given Indian market focus)</p> <ul style="list-style-type: none"> <li>• Use <code>next-intl</code> package</li> <li>• URL pattern: <code>/en/learn/...</code> vs <code>/hi/learn/...</code></li> <li>• Content translated into Hindi, Tamil, Telugu</li> <li>• Admin panel allows adding translated content per topic</li> <li>• Huge SEO opportunity for regional engineering keywords</li> </ul> <hr/> <p><b>Progressive Web App (PWA)</b></p> <p><b>When:</b> Mobile traffic exceeds 30%.</p> <p>Add service worker:</p> <ul style="list-style-type: none"> <li>• <b>Offline reading:</b> Previously visited topics cached and readable without internet</li> <li>• <b>Home screen install:</b> Prompt users to add as app icon</li> <li>• <b>Background sync:</b> Mark topics complete offline → syncs when reconnected</li> </ul> <hr/> <p><b>React Native Mobile App</b></p> <p><b>When:</b> PWA feels insufficient and mobile DAU is significant.</p> <p><b>Stack:</b> React Native + Expo</p> <ul style="list-style-type: none"> <li>• Shares all API calls with the web platform</li> <li>• Native push notifications for POTD reminders, streak alerts, new content</li> <li>• Offline content caching</li> <li>• Biometric login (Face ID / Fingerprint)</li> <li>• Monaco or CodeMirror mobile for the code editor</li> </ul> <hr/> | Test | Variant A | Variant B | Price point | ₹199/month | ₹299/month | CTA copy | "Unlock All Content" | "Start Learning" | Page layout | Comparison table | Feature list |
| Test        | Variant A            | Variant B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |      |           |           |             |            |            |          |                      |                  |             |                  |              |
| Price point | ₹199/month           | ₹299/month                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |      |           |           |             |            |            |          |                      |                  |             |                  |              |
| CTA copy    | "Unlock All Content" | "Start Learning"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |      |           |           |             |            |            |          |                      |                  |             |                  |              |
| Page layout | Comparison table     | Feature list                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |      |           |           |             |            |            |          |                      |                  |             |                  |              |

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### 16.8 Technology Upgrade Path

| MVP (Now)                      |   | Scale (Later)                                                                                      |
|--------------------------------|---|----------------------------------------------------------------------------------------------------|
| MongoDB Atlas M0 (free)        | → | M10+ cluster (\$57/month) at 10K+ users                                                            |
| Node.js in-memory rate limiter | → | Redis (Upstash serverless) for distributed rate limiting across multiple Vercel function instances |
| MongoDB text search            | → | Algolia or Meilisearch for better relevance                                                        |
| localStorage auth              | → | Full NextAuth.js with DB adapter (persistent sessions)                                             |
| Static /public uploads         | → | S3 + CloudFront or Cloudinary for all images at scale                                              |
| Vercel Hobby                   | → | Vercel Pro (\$20/month) for higher function limits                                                 |
| Manual sitemap trigger         | → | Cron job (Vercel Cron) to auto-rebuild sitemap nightly                                             |

End of System Design Document — EngineerTutorial Platform v1.0 Includes: Functional Requi  
API Design · Workflows · Future Roadmap